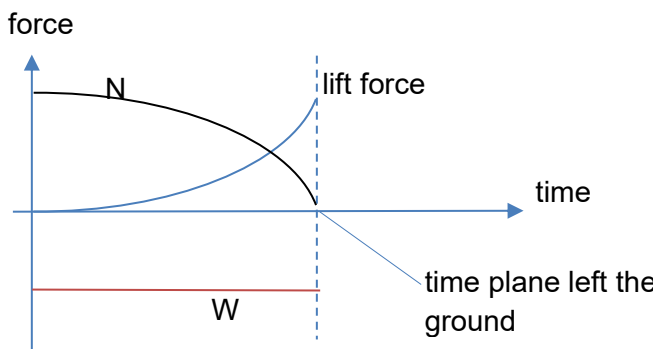


Questions		Answers	Marks
9	(a)	$\frac{(18\,000 - 3\,000)\text{ kg}}{5\text{ kg km}^{-1}} = 3\,000\text{ km}$	B1
	(b)	$\frac{3\,000\text{ kg}}{5\text{ kg km}^{-1}} = 600\text{ km}$	B1
	(c)	$42\,000 + 18\,000 + 15\,000 = 75\,000\text{ kg}$ total mass = mass of plane + mass of fuel (at full capacity) + mass of 150 passengers	B1
	(d)	 Fig. 9.2 vertical forces are balanced lift force + normal contact force + weight = 0	B1; B1
	(e)	Using $v^2 = u^2 + 2as$, $u = 0$, gives $75^2 = 0 + 2(a)(1500)$ $a = 1.88\text{ m s}^{-2}$	M1 A1
	(f)	Using $F = ma$ gives $F = 1.875 \times 75\,000 = 140\,625 = 141\,000\text{ N}$	A1
	(g) (i)	Since $v^2 = u^2 + 2as$, $u = 0$ gives $a = \frac{v^2}{2s}$ and $F_{net} = ma$ So $\frac{F_{net}}{m} = \frac{v^2}{2s}$	M1
		Since F_{net} , v^2 are constants, $m \propto s$, $\frac{m}{75\,000} = \frac{1200}{1500}$, $m = 60\,000\text{ kg}$ alternative,	A0
		$75^2 = 0 + 2(a)(1200)$ gives $a = 2.3438\text{ m s}^{-2}$	M1
		$140\,625 = m \times 2.3438$ gives $m = 60\,000\text{ kg}$	A0

		(ii)	<table><tr><td>Length of take-off</td><td>1500 m</td><td>1200 m</td><td>1200 m</td><td>1200 m</td></tr><tr><td>Number of passengers</td><td>150</td><td>130</td><td>110</td><td>90</td></tr><tr><td>Mass of aeroplane / kg</td><td colspan="4">42 000</td></tr><tr><td>Mass of passengers / kg</td><td>15 000</td><td>13 000</td><td>11 000</td><td>9 000</td></tr><tr><td>Total mass of fuel /kg</td><td>18 000</td><td>5 000</td><td>7 000</td><td>9 000</td></tr><tr><td>Total mass / kg</td><td>75 000</td><td>60 000</td><td>60 000</td><td>60 000</td></tr><tr><td>Mass of fuel in reserve / kg</td><td colspan="4">3 000</td></tr><tr><td>Usable mass of fuel /kg</td><td>15 000</td><td>2 000</td><td>4 000</td><td>6 000</td></tr><tr><td>Maximum safe distance / km</td><td>3000</td><td>400</td><td>800</td><td>1 200</td></tr></table> C1: Usable mass of Fuel A1: last row of column 2, 3, 4 Total mass = mass of plane + mass of fuel + mass of passengers Eg column 2: 60 000 = 42 000 + (mass of usable fuel + 3000) + (130 x 100) - max safe distance = mass of usable fuel / 5 kg per km	Length of take-off	1500 m	1200 m	1200 m	1200 m	Number of passengers	150	130	110	90	Mass of aeroplane / kg	42 000				Mass of passengers / kg	15 000	13 000	11 000	9 000	Total mass of fuel /kg	18 000	5 000	7 000	9 000	Total mass / kg	75 000	60 000	60 000	60 000	Mass of fuel in reserve / kg	3 000				Usable mass of fuel /kg	15 000	2 000	4 000	6 000	Maximum safe distance / km	3000	400	800	1 200	C1 A1
Length of take-off	1500 m	1200 m	1200 m	1200 m																																													
Number of passengers	150	130	110	90																																													
Mass of aeroplane / kg	42 000																																																
Mass of passengers / kg	15 000	13 000	11 000	9 000																																													
Total mass of fuel /kg	18 000	5 000	7 000	9 000																																													
Total mass / kg	75 000	60 000	60 000	60 000																																													
Mass of fuel in reserve / kg	3 000																																																
Usable mass of fuel /kg	15 000	2 000	4 000	6 000																																													
Maximum safe distance / km	3000	400	800	1 200																																													
	(h)	Any reasonable answer explained with sound physics concepts. Example Examples: B1: <u>Kinetic energy increases with mass</u> ($E_k = \frac{1}{2}mv^2$) B1: So more energy is needed to be dissipated / more stress force on landing gear.	B1 B1																																														
		or B1: With <u>large mass, more reaction force</u> by ground on plane / and by plane on ground. (Newton's third law) B1: More impact load at landing, increases probability of damage on landing gear (force by ground on plane) / increase probability of damage on runway (force by plane on ground)	B1 B1																																														